

MASS-ENG-021

Dispatch Engineering Specification

Version 1.0

Engineering Objective

Dispatch implements the constitutional capability responsible for coordinating enterprise field execution. Dispatch transforms operational work into scheduled, assigned, monitored, and completed field activities while preserving constitutional ownership boundaries.

Engineering Preface

Dispatch is not enterprise operations. Operations determines what work must be performed; Dispatch determines when, where, by whom, and in what sequence field work is executed. This capability is enterprise-wide and shall support technicians, inspectors, case workers, delivery personnel, volunteers, service professionals, and future field roles without redesign.

Purpose & Scope

Dispatch governs scheduling, assignment, routing, workload balancing, resource availability, field status, escalation, and execution visibility. Out of Scope: relationship ownership, customer experience stewardship, growth strategy, financial processing, and communications strategy.

Architecture

Core Components: Dispatch Service, Scheduling Service, Assignment Service, Route Optimization Service, Resource Registry, Availability Service, Capacity Service, Status Service, Escalation Service, Dispatch Intelligence Service.

Engineering Requirements

Functional: schedule work, assign resources, optimize routes, balance workloads, monitor execution, manage exceptions, recommend reassignment, provide real-time operational visibility.

Non-Functional: scalable, resilient, observable, auditable, secure, constitutionally compliant.

Dependencies & Responsibilities

Requires Enterprise Core, Identity, Security. Uses Workflow, Knowledge, Documents, AI Orchestration, Notifications, Observability, Persistence, Error, Configuration, API, Integration, Relationship Command, Customer Experience, Growth, and Operations. Provides Dispatch Intelligence, Scheduling, Assignment Coordination, Resource Visibility, Field Execution Status. Owns field coordination; does not own operations, finance, communications, relationships, or customer experience.

Interfaces, Example Flow & Definition of Done

Interfaces: ScheduleWork(), AssignResource(), OptimizeRoute(), UpdateStatus(), ReassignWork(), CloseAssignment(). Example Flow: Operations creates work -> Dispatch schedules and assigns resources -> Communications notifies stakeholders -> Field resource executes work -> Status updates published -> Operations records completion -> Finance processes value. Definition of Done: Dispatch consistently coordinates enterprise field execution with real-time visibility, intelligent scheduling, and measurable operational performance.